

Workshop 8

Problem 1

Keywords: Derivatives of Inverse Functions, Tangent Lines to Inverse Functions

Let $f(x) = x^3 + x + 2$. The function f is invertible, but we do not know how to determine $f^{-1}(x)$ explicitly.

- (a) Find $f(1)$ and $f'(1)$.
- (b) Find $f^{-1}(4)$ and $(f^{-1})'(4)$.
- (c) Find an equation of the tangent line to $y = f^{-1}(x)$ at $x = 4$.
- (d) Find all the points on the graph of $y = f^{-1}(x)$ where the tangent line has slope $\frac{3}{4}$.

(Hint: if (a, b) is a point of the graph of f where the tangent has slope m , then (b, a) is a point on the graph of f^{-1} where the tangent has slope $\frac{1}{m}$.)

Problem 2

Keywords: Logarithmic Differentiation

Let $f(x) = x^{7x}$. Alice uses the fact that $\frac{d}{dx}x^n = nx^{n-1}$ and decides

$$f'(x) = 7x \cdot x^{7x-1} = 7x^{7x}$$

Beatriz uses the fact that $\frac{d}{dx}a^x = \ln(a)a^x$ together with the chain rule and concludes that

$$f'(x) = x^{7x} \cdot 7 \ln(x).$$

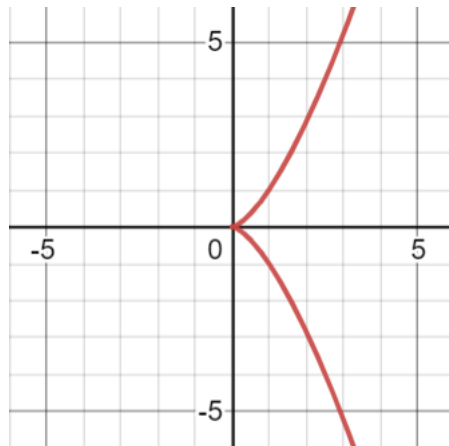
- (a) Explain in 1-2 English sentences why Alice and Beatriz are wrong.
- (b) Find $f'(x)$.
- (c) Find the equation(s) of the horizontal tangent line(s) to the graph of f .

Problem 3

Keywords: Related Rates

A particle is moving on the curve $y^2 = x^3$ (pictured below). When $y = -8$, the y -coordinate (measured in feet) of the particle is increasing at a rate of 3 ft/s. What is the rate of change of the angle of inclination of the line joining the particle and the origin when $y = -8$?

Be sure you define any variables used, draw an appropriate picture (or complete the one below), use calculus to justify your answer, and report your answer in a sentence with appropriate units.



Optional Problem *(for fun, no need to submit)*

Let $g(x) = (2x + 1)^{\left(\csc^{-1}(x^2)^{3x \tan(x)}\right)}$. Find $g'(x)$.